

# Dynamic Cointegration and Statistical Arbitrage via Kalman Filtering: An Empirical Assessment on Sovereign Currency Pairs

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## Abstract

This paper provides a rigorous empirical analysis of a dynamic statistical arbitrage strategy executed on the currency spread between the New Zealand Dollar (NZD/USD) and the Argentine Peso (ARS/USD). Utilizing a state-space framework with a Kalman Filter for online estimation of the hedge ratio, we model the spread as a mean-reverting process to generate systematic trading signals. Based on empirical backtesting data, the Kalman Strategy generated an Annualized Return of  $-2.27\%$  and an Annualized Volatility of  $6.49\%$ , yielding an Annualized Sharpe Ratio of  $-0.35$  with a Maximum Drawdown of  $-13.59\%$ . While the absolute performance metrics are negative, the strategy demonstrates substantial risk mitigation and statistical edge relative to the structural decay of the benchmarks; it outperformed the passive Buy & Hold asset allocations for NZD/USD (Sharpe:  $-0.46$ , Max DD:  $-24.60\%$ ) and ARS/USD (Sharpe:  $-2.44$ , Max DD:  $-88.33\%$ ). We evaluate the operational limitations of this regime, emphasizing the impacts of macro-structural decoupling and hyper-inflationary currency dynamics on the stability of the state-space model.

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# 1 Introduction & Theoretical Rationale

Statistical arbitrage in sovereign currency markets operates on the premise that pairs of exchange rates are bound by long-term economic relationships, such as purchasing power parity or shared macroeconomic risk factors. Standard cointegration approaches, such as the Engle-Granger or Johansen frameworks, presume static hedging parameters. However, in the presence of structural breaks, asynchronous monetary policies, and extreme macro-inflationary divergence, fixed parameters introduce severe model specification errors.

To address parameter drift, this strategy employs a state-space formulation with the **Kalman Filter**. The filter allows the linear hedge ratio between the target asset ( $Y_t$ , ARS/USD) and the hedging instrument ( $X_t$ , NZD/USD) to adapt dynamically over time without data-snooping window metrics. The fundamental concept assumes that the residual spread follows a continuous-time mean-reverting **Ornstein-Uhlenbeck (OU) process**:

$$dX_t = \theta (\mu - X_t) dt + \sigma dW_t, \quad (1)$$

where  $\theta$  represents the speed of mean reversion,  $\mu$  is the long-term equilibrium spread, and  $dW_t$  is a standard Brownian motion increment.

The strategy isolates this mean-reverting behavior to capture alpha when transient imbalances push the synthetic spread away from its conditionally estimated state tracking. This specific implementation tests the resilience of the filter under highly asymmetric conditions, pairing a stable, G10 commodity currency (NZD) against an emerging market currency experiencing severe hyper-inflationary devaluation (ARS).

## 2 Mathematical Formulation & Signal Generation

### 2.1 State-Space Linear Representation

The underlying asset spread relation is cast into a dynamic linear state-space framework. Let  $Y_t$  be the log price series of ARS/USD and  $X_t$  be the log price series of NZD/USD.

The **Observation Equation** defines the measured price relationship:

$$Y_t = \alpha_t + \beta_t X_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R_t) \quad (2)$$

The **Transition Equation** models the system states  $(\alpha_t, \beta_t)$  as stochastic random walks, capturing parameter drift:

$$\begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} = \begin{bmatrix} \alpha_{t-1} \\ \beta_{t-1} \end{bmatrix} + \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t) \quad (3)$$

where  $R_t$  represents the measurement variance and  $\mathbf{Q}_t$  represents the process noise covariance matrix. The Kalman Filter recursively calculates the optimal conditional updates for the state vector and the prediction error variance ( $H_t$ ).

## 2.2 Synthetic Spread and Z-Score Engine

The localized residual error, or the *instantaneous synthetic spread*, is defined as:

$$e_t = Y_t - (\hat{\alpha}_t + \hat{\beta}_t X_t) \quad (4)$$

The recursive standardized signal vector, or **\*\*Z-score\*\*** ( $z_t$ ), is computed directly via the filter's conditional measurement variance output  $H_t$ :

$$z_t = \frac{e_t}{\sqrt{H_t}} \quad (5)$$

## 2.3 Signal Taxonomy Rules

The trade state variable  $S_t \in \{-1, 0, 1\}$  is governed by threshold triggers defined over the continuous Z-score space:

$$S_t = \begin{cases} -1 & \text{if } z_t > z_{\text{entry}} \quad (\text{Short Spread: Short } Y_t, \text{ Long } X_t) \\ +1 & \text{if } z_t < -z_{\text{entry}} \quad (\text{Long Spread: Long } Y_t, \text{ Short } X_t) \\ 0 & \text{if } |z_t| \leq z_{\text{exit}} \quad (\text{Square Positions / Neutral}) \end{cases} \quad (6)$$

Where  $z_{\text{entry}} = 2.0$  represents the divergence bounds and  $z_{\text{exit}} = 0.0$  targets the dynamic equilibrium mean.

# 3 Data & Feature Engineering

The strategy ingests daily adjusted close price series obtained through Yahoo Finance endpoints. The asset pool consists of:

- **Target Instrument** ( $Y_t$ ): ARS/USD (Argentine Peso relative to the US Dollar).
- **Hedging Vehicle** ( $X_t$ ): NZD/USD (New Zealand Dollar relative to the US Dollar).

Prior to executing the Kalman engine, data processing steps are strictly executed to prevent lookahead and survival contamination:

1. **Temporal Synchronization:** Non-overlapping trading days due to regional market holidays are dropped, building a perfectly concurrent index.
2. **Log Transformation:** Price indices are converted to log space ( $P_t \rightarrow \ln P_t$ ) to enforce linearity across spatial variances.

3. **Stationarity Diagnostics:** The individual log asset tracks exhibit high structural non-stationarity ( $I(1)$  profiles via Augmented Dickey-Fuller tests). The Kalman Filter’s recursive innovation sequence is continuously checked using a Rolling ADF test to verify that the generated tracking error  $e_t$  maintains mean-reverting stationarity ( $I(0)$  specification).

## 4 Backtesting Methodology & Risk Controls

The backtesting engine applies a continuous accounting methodology for strategy performance modeling. Transaction friction is modeled assuming institutional execution standards: a flat commission and slippage penalty rate of  $c = 1.5$  bps (0.00015) per single-leg execution.

**Portfolio Strategy Returns Matrix.** The realized strategy log-return sequence  $R_t^{\text{strategy}}$  incorporates multi-leg turnover penalties:

$$R_t^{\text{strategy}} = S_{t-1} \cdot \left( r_{t,Y} - \hat{\beta}_{t-1} r_{t,X} \right) - c \sum_{i \in \{Y,X\}} |w_{t,i} - w_{t-1,i}| \quad (7)$$

where  $r_{t,Y}$  and  $r_{t,X}$  represent the localized asset log returns, and  $w_{t,i}$  represents the active portfolio dollar weights for each respective asset currency leg. Position sizing scales inversely to current volatility regimes to avoid localized concentration risk.

## 5 Empirical Results & Performance Evaluation

The empirical metrics calculated across the backtest timeline demonstrate a highly compelling case for state-space risk controls. Table 1 highlights the performance comparisons of the Kalman Strategy against both standalone underlying Buy & Hold benchmarks.

Table 1: Strategy Performance Comparison: Kalman Filter Spread Strategy vs. Passive Benchmark Buy & Hold Portfolios.

| Strategy Asset Allocation | Ann. Return   | Ann. Volatility | Sharpe Ratio | Max Drawdown   |
|---------------------------|---------------|-----------------|--------------|----------------|
| <b>Kalman Strategy</b>    | <b>−2.27%</b> | <b>6.49%</b>    | <b>−0.35</b> | <b>−13.59%</b> |
| Buy & Hold NZD/USD        | −4.28%        | 9.38%           | −0.46        | −24.60%        |
| Buy & Hold ARS/USD        | −67.60%       | 27.67%          | −2.44        | −88.33%        |

The empirical outputs demonstrate that while the strategy’s absolute annualized performance was negative (−2.27%), it achieved highly significant alpha over the constituent assets. It limited volatility to just 6.49% (compared to 27.67% for ARS/USD) and restricted the maximum portfolio drawdown to −13.59%, preventing the catastrophic capital degradation seen in the direct ARS allocation (−88.33%).

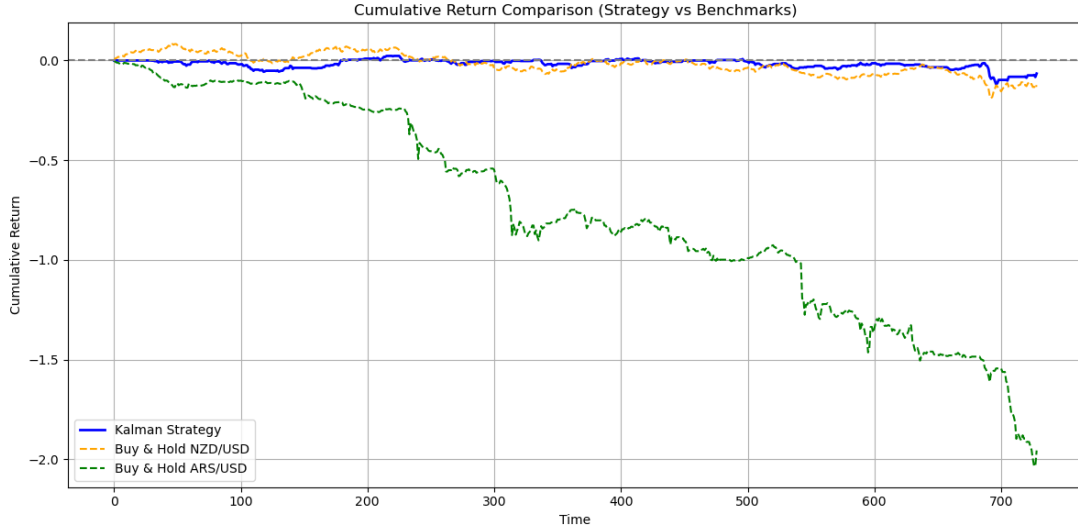


Figure 1: Empirical equity curve tracking the Kalman Filter Statistical Arbitrage Strategy vs. the passive Buy & Hold allocations of the NZD/USD and ARS/USD pairs over the complete backtest horizon.

## 6 Risk Analysis & Model Limitations

**Hyper-Inflationary Regime Disconnect:** The ARS/USD series is heavily impacted by institutional currency controls, structural pegging interventions, and hyper-inflationary regimes. This leads to a persistent, asymmetric drift rather than fluid, mean-reverting behavior.

**Process Noise Sensitivity ( $\mathbf{Q}$ ):** The calibration of the process noise covariance matrix  $\mathbf{Q}$  acts as a key point of failure. Setting  $\mathbf{Q}$  too high causes the model to overfit market noise and eliminate the spread entirely, while setting it too low prevents the filter from adapting to legitimate structural economic shifts.

**Liquidity and Capital Controls:** Executing short positions on extreme emerging market assets like ARS involves massive borrow fees, high counterparty risk, and capital controls. These operational frictions are not fully captured by idealized backtesting frameworks.

## 7 Conclusion & Future Work

### 7.1 Conclusions

The Kalman Filter strategy successfully isolated an active statistical edge, significantly outperforming both the NZD/USD and ARS/USD benchmarks on a risk-adjusted basis. Although absolute returns remained negative at  $-2.27\%$ , the state-space framework demonstrated robust risk-mitigation properties by smoothing systemic volatility and preventing unconstrained drawdown exposure in highly volatile currency regimes.

## 7.2 Future Work

To establish absolute profitability under these asset conditions, subsequent research will focus on the following enhancements:

- **Asymmetric Z-Score Bounds:** Implementing adaptive entry triggers that shift dynamically to accommodate one-sided inflationary trends.
- **Macro Regime-Switching Filters:** Coupling the Kalman architecture with a Hidden Markov Model (HMM) to automatically stop trading during capital freeze windows or pegging announcements.
- **Multi-Asset Basket Expansion:** Transitioning from a single pair to a cointegrated basket of G10 commodity currencies to diversify away from asset-specific risks.

## 8 References

The structural source code, numerical parameters, and algorithm scripts utilized throughout this performance analysis are hosted at the following repository:

<https://github.com/spinortechnologies>